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A cross-sectional view of a semiconductor device. The device is formed on a substrate 130. The device includes a first layer 100 and a second layer 105. The first layer 100 includes a first conductive layer 110 and a second conductive layer 115. The second layer 105 includes a third conductive layer 120 and a fourth conductive layer 125. The device is formed on a substrate 130. Various regions are labeled with reference numerals: 155, 165, 175, 135, 110, 120, 115, 125, 130, 150, 145, 140, 160, 170, and 105.

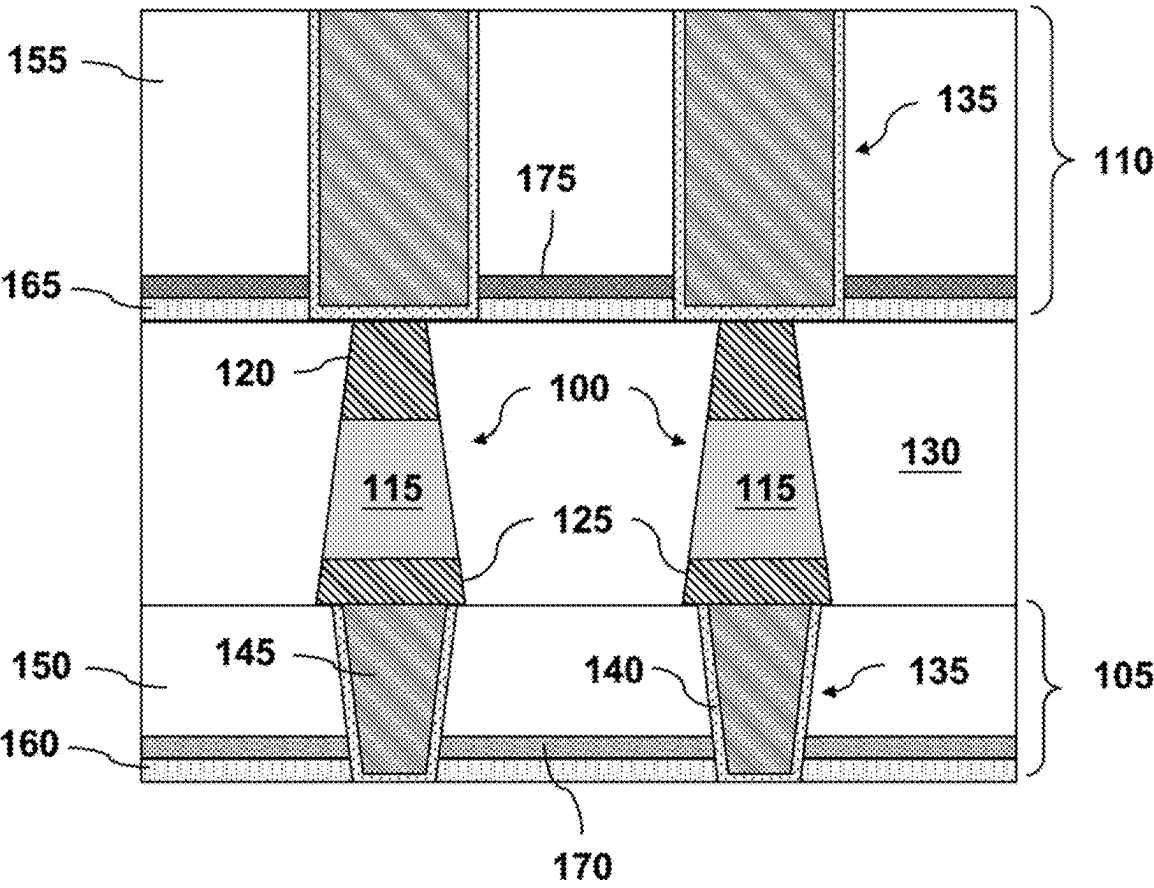


FIG. 1

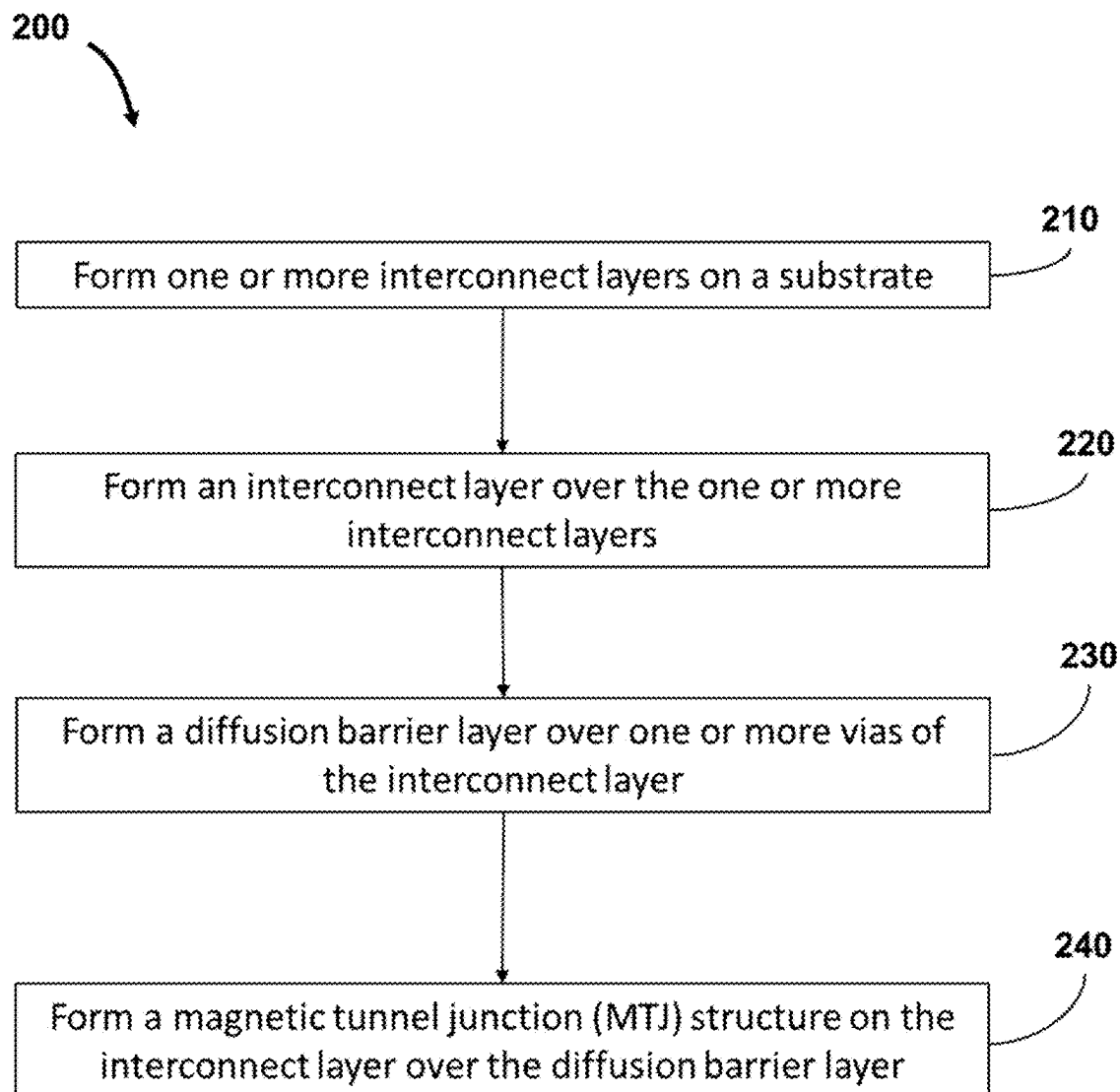


FIG. 2

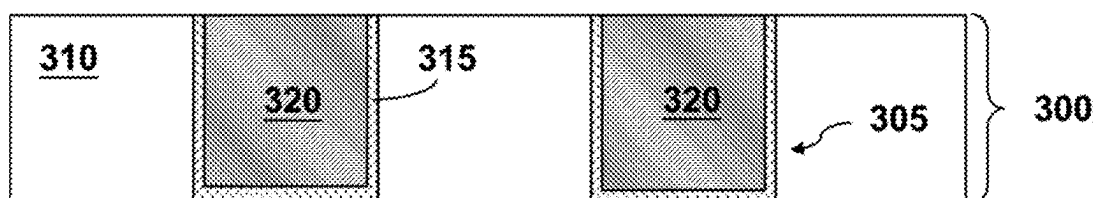


FIG. 3

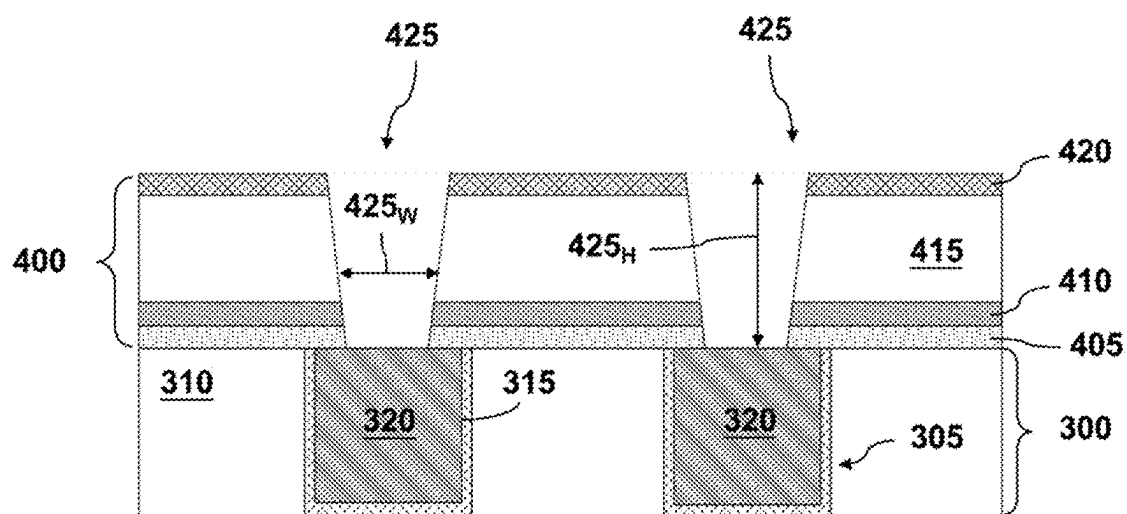


FIG. 4

FIG. 5

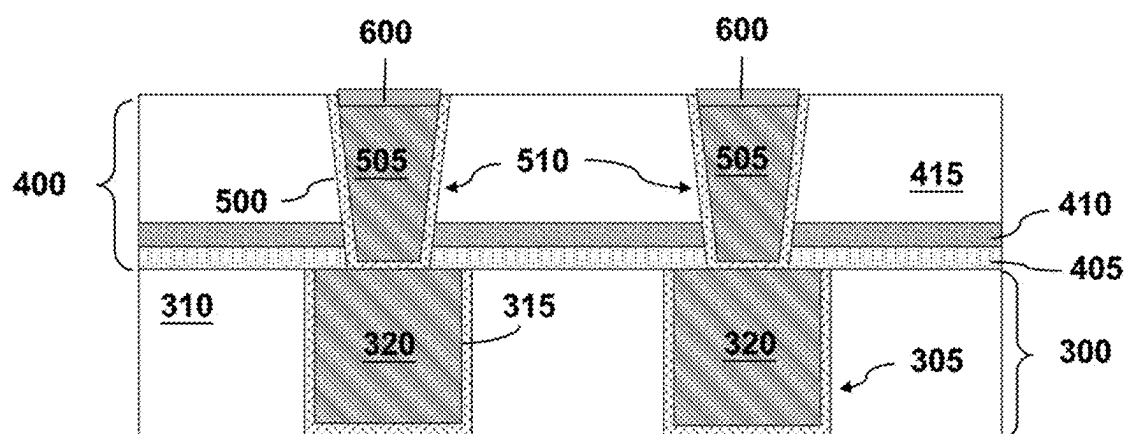


FIG. 6

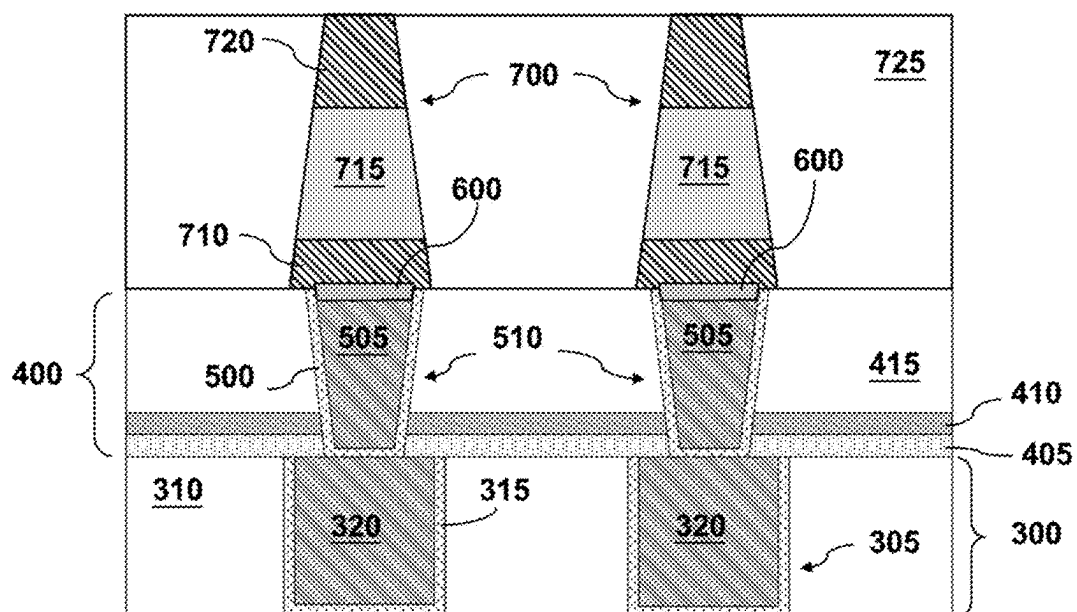


FIG. 7

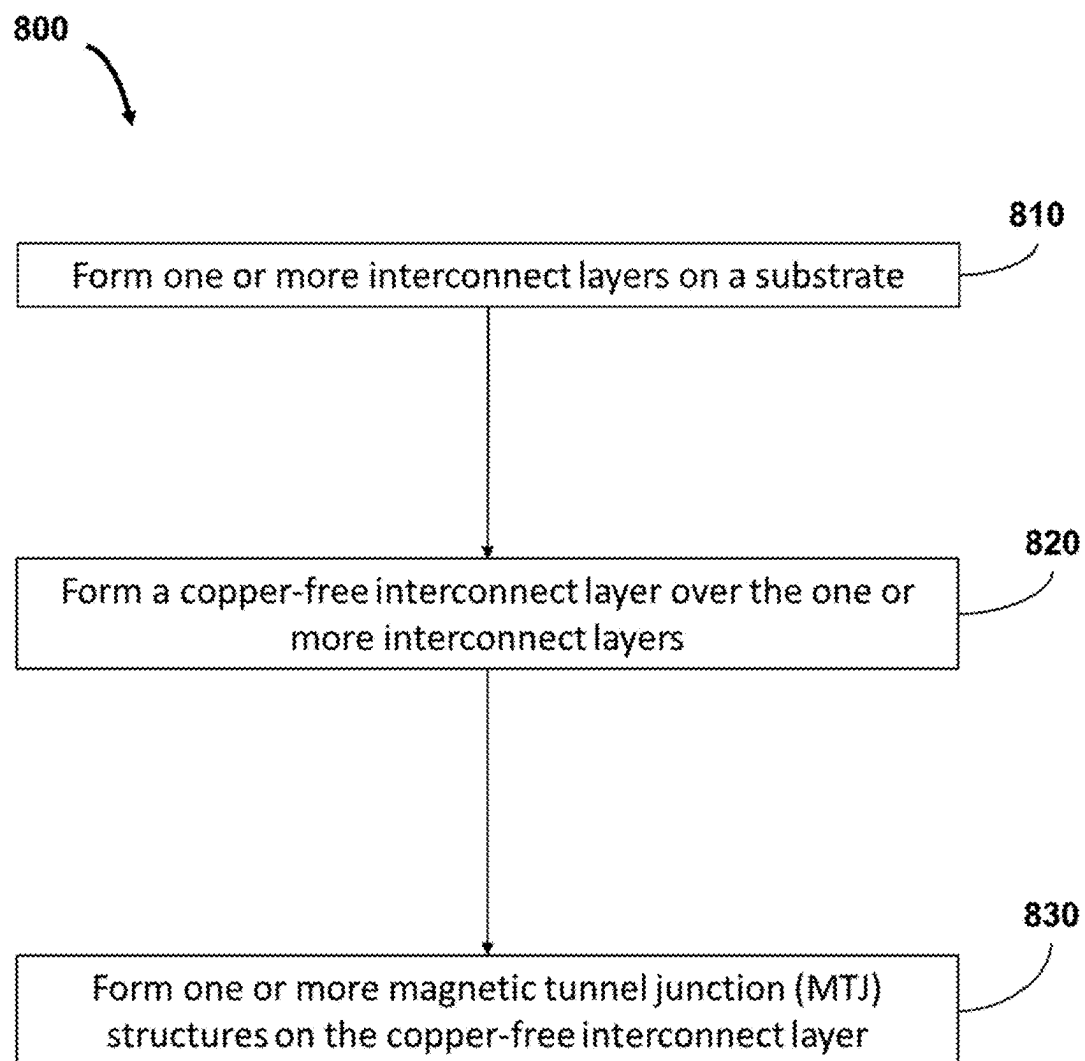


FIG. 8

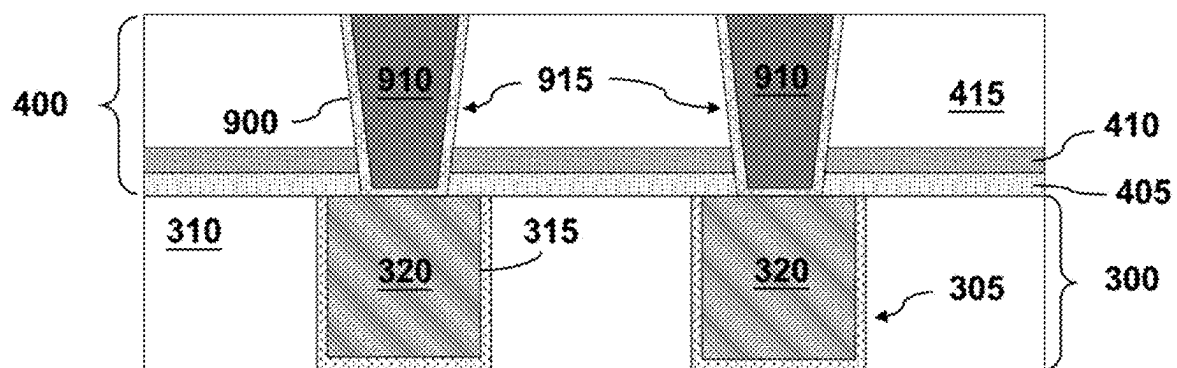


FIG. 9

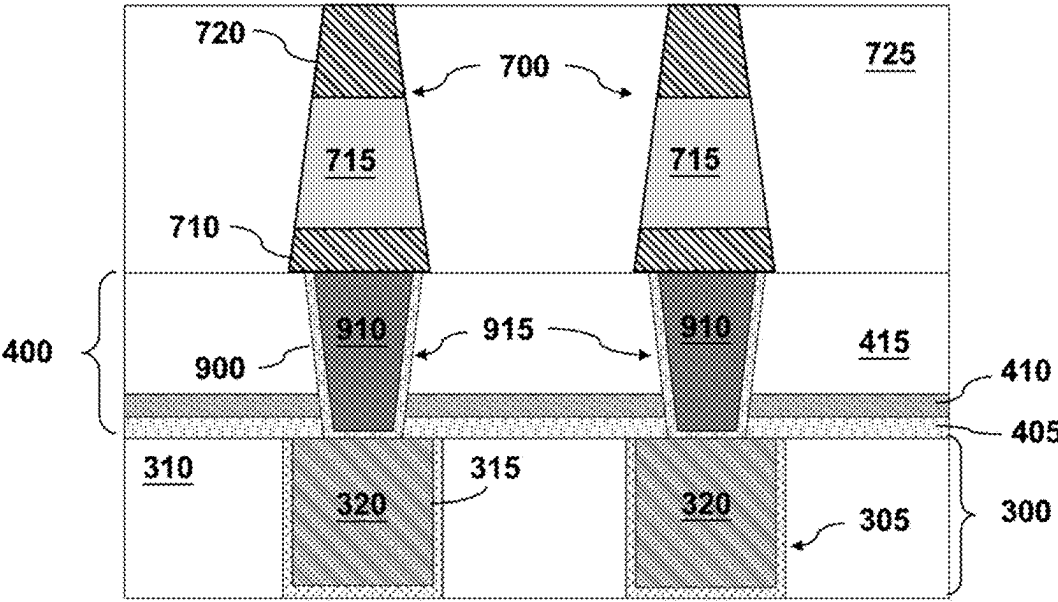


FIG. 10

DIFFUSION LAYER FOR MAGNETIC TUNNEL JUNCTIONS

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation of U.S. patent application Ser. No. 18/148,829, titled “Diffusion Barrier Layer on Interconnection Vias for Magnetic Tunnel Junctions,” filed Dec. 30, 2022, which is a continuation of U.S. patent application Ser. No. 17/334,536, titled “Diffusion Layer for Magnetic Tunnel Junctions,” filed May 28, 2021, which is a divisional of U.S. patent application Ser. No. 16/210,226, titled “Diffusion Layer for Magnetic Tunnel Junctions,” filed Dec. 5, 2018, which claims the benefit of U.S. Provisional Patent Application No. 62/690,638, titled “Diffusion Layer for Magnetic Tunnel Junctions,” filed Jun. 27, 2018, each of which is incorporated by reference in its entirety.

BACKGROUND

[0002] In integrated circuits (ICs), magnetic tunneling junctions (MTJs) are an integral part of magnetic random access memories (MRAMs). The MTJ structures can be formed in the back end of the line (BEOL) between layers of interconnects (e.g., lines and vias) that include a metal (e.g., copper) or metal alloy (e.g., copper alloy). Diffusion of the metal or metal alloy from the interconnect layers to the MTJ structures can disrupt the MRAMs’ operation.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the common practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 is a cross-sectional view of a magnetic tunnel junction structure that is formed between two interconnect layers, according to some embodiments.

[0005] FIG. 2 is a flowchart of a fabrication method for forming a capping layer between an interconnect layer and the bottom electrodes of magnetic tunnel junction structures, according to some embodiments.

[0006] FIG. 3 is a cross-sectional view of an interconnect layer, according to some embodiments.

[0007] FIG. 4 is a cross-sectional view of an interconnect layer after the formation of via openings, according to some embodiments.

[0008] FIG. 5 is a cross-sectional view of an interconnect layer after the formation of a chemical mechanical planarization (CMP) operation, according to some embodiments.

[0009] FIG. 6 is a cross-sectional view of two interconnect layers with the top interconnect layer having a diffusion barrier layer (capping layer) selectively formed over its vias, according to some embodiments.

[0010] FIG. 7 is a cross-sectional view of magnetic tunnel junction (MTJ) structures over an interconnect layer with a capping layer formed between the bottom electrode of the MTJ structures and the vias of the interconnect layer, according to some embodiments.

[0011] FIG. 8 is a flowchart of a fabrication method for forming a copper-free interconnect layer below magnetic tunnel junction structures, according to some embodiments.

[0012] FIG. 9 is a cross-sectional view of an interconnect layer after the formation of copper-free metal vias, according to some embodiments.

[0013] FIG. 10 is a cross-sectional view of magnetic tunnel junction (MTJ) structures formed over a copper-free interconnect layer, according to some embodiments.

DETAILED DESCRIPTION

[0014] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0015] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0016] The term “about” as used herein indicates the value of a given quantity that can vary based on a particular technology node associated with the subject semiconductor device. In some embodiments, based on the particular technology node, the term “about” can indicate a value of a given quantity that varies within, for example, 5-30% of the value (e.g., $\pm 5\%$, $\pm 10\%$, $\pm 20\%$, or $\pm 30\%$ of the value).

[0017] The term “nominal” as used herein refers to a desired, or target, value of a characteristic or parameter for a component or a process operation, set during the design phase of a product or a process, together with a range of values above and/or below the desired value. The range of values can be due to slight variations in manufacturing processes or tolerances. Unless defined otherwise, technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art to which this disclosure belongs.

[0018] In integrated circuits (ICs), magnetic tunneling junctions (MTJs) are an integral part of magnetic random access memories (MRAMs). The MTJ structures can be formed in the back end of the line (BEOL) between layers of interconnects filled with metal (e.g., copper (Cu)) or a metal alloy (e.g., copper alloy (Cu-alloy)). Out-diffusion of the metal or the metal alloy from neighboring interconnect layers into the MTJ structures can disrupt the MRAMs’ operation and is therefore undesirable.

[0019] The embodiments described herein are directed to exemplary interconnect fabrication methods that can prevent metal (e.g., Cu) out-diffusion towards neighboring MTJ structures. For example, in some embodiments, a method is described for the formation of a cobalt (Co) or ruthenium (Ru) diffusion barrier layer between the Cu interconnects and the MTJ structures. The diffusion barrier layer can be selectively formed over Cu interconnects connected to bottom electrodes of the MTJ structures. As a result, the diffusion barrier layer can prevent out-diffusion of Cu from the interconnect layers. In another embodiment, a method is described that forms a Cu-free interconnect layer using a tungsten (W) metallization. W, unlike Cu, is not mobile and therefore does not out-diffuse to neighboring MTJ structures.

[0020] As discussed above, MTJ structures can be formed between BEOL interconnect layers. For example, one or more MTJ structures can be formed in an interlayer dielectric (ILD) between two interconnect layers. A cross-sectional view of exemplary MTJ structures **100** between two interconnect layers **105** and **110** is shown in FIG. 1. MTJ structure **100** is a multilayer structure that includes MTJ stack **115**, a top electrode **120**, and a bottom electrode **125**. MTJ stack **115** can further include a non-conductive layer (not shown) disposed between two ferromagnetic layers (not shown). In some embodiments, the non-conductive layer can include magnesium oxide (MgO), aluminum oxide (AlO_x), aluminum oxynitride (AlON), or combinations thereof. According to some embodiments, the non-conductive layer can be deposited by a physical vapor deposition (PVD) technique. Alternatively, the non-conductive layer can be deposited by other deposition techniques, such as chemical vapor deposition (CVD), plasma-enhanced CVD (PECVD), atomic layer deposition (ALD), plasma-enhanced ALD (PEALD), or any other suitable deposition technique.

[0021] In some embodiments, the ferromagnetic layers can be metal stacks with one or more layers that include any combination of iron (Fe), cobalt (Co), ruthenium (Ru), and magnesium (Mg). Further, the ferromagnetic layers can be deposited by PVD, PEVD, CVD, PECVD, ALD, PEALD, or any other suitable deposition method. In some embodiments, the thickness of MTJ stack **115** can range from about 100 Å to about 400 Å.

[0022] Each of top and bottom electrodes **120** and **125** respectively are in electrical and physical contact with the ferromagnetic layers of MTJ stack **115**. According to some embodiments, top electrode **120** can include tantalum (Ta), tantalum nitride (Ta₂N), titanium nitride (TiN), or combinations thereof. For example, top electrode **120** can be a stack that includes a bottom TiN layer and a top TaN layer, which can be deposited by CVD or PVD. Bottom electrode **125** can include TiN, TaN, Ru, or combinations thereof and can be deposited by CVD or PVD. In some embodiments, top and bottom electrodes **120** and **125** can each have a thickness between about 300 Å and about 800 Å.

[0023] MTJ structure **100** can also include additional layers which are not shown in FIG. 1 for simplicity. Such layers can be for example one or more capping layers, or spacers, that electrically isolate MTJ structure **100**. By way of example and not limitation, the capping layers, which are not shown in FIG. 1, can include a stack of materials, such as silicon nitride (SiN), silicon oxide (SiO₂), silicon carbon nitride (SiCN), or combinations thereof. In some embodiments, the capping layers can be deposited by CVD or ALD

at a deposition temperature between about 140° C. and about 250° C. Further, the thickness of the capping layers can range from about 50 Å to about 3000 Å, according to some embodiments.

[0024] As shown in FIG. 1, MTJ structure **100** is embedded in ILD **130**. In some embodiments, ILD **130** can be a low-k dielectric material with a dielectric constant (e.g., k-value) below 3.9 (e.g., 3.6), silicon nitride, silicon oxide, silicon oxynitride, fluorine-doped silicate glass (FSG), or an undoped oxide (UDOX). In some embodiments, ILD **130** can be a stack of dielectrics such as a low-k dielectric and another dielectric. The stack of dielectrics can be, for example, (i) a low-k dielectric (e.g., carbon doped silicon oxide) and a silicon carbide with nitrogen doping; (ii) a low-k dielectric and a silicon carbide with oxygen doping; (iii) a low-k dielectric with silicon nitride; or (iv) a low-k dielectric with silicon oxide. Further, ILD **130** can be deposited by an atmospheric CVD (APCVD) process, a high-density plasma CVD (HDPCVD) process, or a PECVD process.

[0025] As shown in FIG. 1, top electrode **120** can be connected to top interconnect layer **110**. Bottom electrode **125** can be connected to a bottom interconnect layer **105**. By way of example and not limitation, bottom and top interconnect layers **105** and **110** can be formed by BEOL fabrication processes based on a Cu metallization scheme. In some embodiments, MTJ structures **100**, which are embedded in ILD **130**, may also be part of an interconnect layer. For example, ILD **130** may include conductive structures (e.g., vias) that electrically connect interconnect layers **105** and **110**. These conductive structures are not shown in FIG. 1 for simplicity. Further, each of bottom and top interconnect layers **105** and **110** can include a network of interconnects such as vertical interconnect access lines (vias) **135** and lateral lines (lines) (not shown in FIG. 1) embedded in an ILD layer **150** and **155** respectively. Vias **135** provide electrical connections between layers and lines provide electrical connections within a layer. Vias **135** and lines can be filled with a metal stack that includes at least a barrier layer **140** and a metal fill **145**. By way of example and not limitation, vias **135** in interconnect layer **105** can have a smaller width compared to interconnect vias **135** in interconnect layer **110**. In some embodiments, bottom interconnect layer **105** can be referred to as “bottom electrode via layer” because it connects bottom electrodes **125** of MTJ structures **100** with lower interconnect layers below interconnect layer **105**, which are not shown in FIG. 1.

[0026] In some embodiments, barrier layer **140** can be a stack of two or more layers and metal fill **145** can be an electroplated metal. For example, barrier layer **140** can include a bottom TaN layer and a top Ta layer deposited by PVD, and metal fill **145** can be electroplated Cu or a Cu-alloy (e.g., copper manganese (CuMn)).

[0027] In some embodiments, ILD **150** can be a low-k dielectric material with a k-value below 3.9 (e.g., 3.6), silicon nitride, silicon oxide, silicon oxynitride, FSG, or UDOX. In some embodiments, ILD **150** can be a stack of dielectrics such as a low-k dielectric and another dielectric. The stack of dielectrics can be, for example, (i) a low-k dielectric (e.g., carbon doped silicon oxide) and a silicon carbide with nitrogen doping; (ii) a low-k dielectric and a silicon carbide with oxygen doping; (iii) a low-k dielectric with silicon nitride; or (iv) a low-k dielectric with silicon

oxide. ILD **150** can be deposited by an APCVD process, a HDPCVD process, or a PECVD process.

[0028] ILD **155** can be a low-k dielectric material with a k-value below 3.9 or a stack of dielectrics such as a low-k dielectric and another dielectric. The stack of dielectrics can be, for example, (i) a low-k dielectric (e.g., carbon doped silicon oxide) and a silicon carbide with nitrogen doping; (ii) a low-k dielectric and a silicon carbide with oxygen doping; (iii) a low-k dielectric with silicon nitride; or (iv) a low-k dielectric with silicon oxide. ILD **155** can be deposited by a HDPCVD process or a PECVD process.

[0029] In some embodiments, bottom interconnect layer **105** can be formed before MTJ structures **100**, and top interconnect layer **110** can be formed after the formation of MTJ structures **100**. In some embodiments, additional MTJ structures **100** can be formed between interconnect layers **105** and **110**. Top and bottom interconnect layers (e.g., **110** and **105** respectively)—along with the one or more MTJ structures, like MTJ structure **100**—can be part of an integrated circuit (IC) structure. The IC structure can include additional layers, not shown in FIG. 1. For example, additional BEOL layers, middle of the line (MOL) layers, and front-end of the line (FEOL) layers can be formed below interconnect layer **105**. By way of example and not limitation, an FEOL layer can include transistors and capacitor structures. An MOL layer can include a network of contacts that connect the transistors and the capacitor structures in the FEOL to the structures in the BEOL layers.

[0030] In some embodiments, interconnect layers **105** and **110** can include additional layers, such as etch stop layers **160** and **165**, and capping layers **170** and **175**. By way of example and not limitation, etch stop layer **160** can be silicon-carbon nitride (SiCN) or aluminum oxide (AlN) with a thickness range between about 10 Å and about 150 Å; etch stop layer **165** can be a SiCN layer with a thickness range between about 100 Å and about 300 Å; capping layer **170** can be aluminum oxide (Al₂O₃) with a thickness between about 10 Å to about 40 Å; and capping layer **175** can be silicon oxide with a thickness between about 100 Å and about 300 Å.

[0031] As shown in FIG. 1, vias **135** of interconnect layer **105** can be in contact with bottom electrodes **125** of MTJ structures **100**. As a result, Cu atoms from vias **135** of interconnect layer **105** can out-diffuse towards bottom electrodes **125** of MTJ structures **100**. By way of example and not limitation, Cu diffusion can occur due to thermal processing of subsequent layers (“thermal diffusion”) or due to the application of an electric field across vias **135** during normal operation. As discussed above, diffusion of Cu atoms into MTJ structures **100** (Cu “poisoning”) can disrupt the MRAM’s normal operation and cause reading errors. For example, diffused Cu atoms can cause leakage across MTJ stack **115** and prevent MTJ structures **100** from storing charge.

[0032] FIG. 2 is a flowchart of an exemplary fabrication method **200** for forming a diffusion barrier layer on vias **135** of interconnect layer **105** prior to the formation of MTJ structures **100**. In some embodiments, the diffusion barrier layer can include Co or another metal, such as Ru. The diffusion barrier layer can be selectively formed over Cu interconnects (e.g., vias **135** of interconnect layer **105**) that contact bottom electrodes **125** of respective MTJ structures **100**. According to some embodiments, the diffusion barrier layer can prevent Cu out-diffusion from vias **135** of inter-

connect layer **105** to the corresponding MTJ structures **100**. Fabrication method **200** may not be limited to the operations described below. Other fabrication operations may be performed between the various operations of fabrication method **200** and are omitted merely for clarity.

[0033] In referring to FIG. 2, exemplary fabrication method **200** begins with operation **210** where one or more interconnects can be formed on a substrate. In some embodiments, the substrate with the one or more interconnect layers can be a partially fabricated wafer in BEOL. Therefore, additional layers can be formed between the substrate and the one or more interconnect layers, such as MOL and FEOL layers. By way of example and not limitation, an FEOL layer can include transistors and capacitor structures, and an MOL layer can provide the electrical connections between the transistors and the capacitor structures in the FEOL layer and the one or more interconnect layers in the BEOL.

[0034] According to some embodiments, FIG. 3 is a cross-sectional view of an exemplary BEOL interconnect layer **300**. Interconnect layer **300** can be, for example, a top interconnect layer from one or more BEOL interconnect layers over a substrate. In the example of FIG. 3, the underlying BEOL interconnect layers, MOL layers, FEOL layers, and the substrate are not shown merely for clarity. In other words, BEOL interconnect layer **300** can be a top layer of a partially fabricated wafer, according to some embodiments. Further, FIG. 3 depicts only a portion of interconnect layer **300**.

[0035] Interconnect layer **300** can include one or more vias **305** and one or more lines (not shown in FIG. 3 for clarity). Vias **305** and lines (not shown) are embedded in ILD **310** and can be filled with a metal stack that includes at least a barrier layer **315** and a metal fill **320**. In some embodiments, barrier layer **315** can be a stack of two or more layers. Metal fill **320** can be a metal or a metal alloy that can be electroplated. By way of example and not limitation, barrier layer **315** can be a TaN/Ta stack deposited by PVD, and metal fill **320** can be electroplated Cu or a Cu-alloy (e.g., CuMn).

[0036] ILD **310** can be a low-k dielectric material or a polymer with a k-value below 3.9 (e.g., **3.6**) or a stack of dielectrics such as a low-k dielectric and another dielectric. By way of example and not limitation, the polymer can be a long carbon chain, a porous polymer, an amorphous polymer, etc. The stack of dielectrics can include, for example, (i) a low-k dielectric (e.g., carbon doped silicon oxide) and a silicon carbide with nitrogen doping; (ii) a low-k dielectric and a silicon carbide with oxygen doping; (iii) a low-k dielectric with silicon nitride; or (iv) a low-k dielectric with silicon oxide. ILD **310** can be deposited by a HDPCVD process or a PECVD process.

[0037] In some embodiments, one or more of interconnect layers **105** and **110** of FIG. 1 are formed in a similar manner as interconnect layer **300**.

[0038] Referring to FIG. 2, fabrication method **200** continues with operation **220** and the formation of another interconnect layer over interconnect layer **300**. For example, as shown in FIG. 4, interconnect layer **400** can be formed over interconnect layer **300**. Interconnect layer **400** can include both vias and lines. Further, the vias of interconnect layer **400** (after a metal fill) can contact vias **305** of interconnect layer **300**. For example purposes, interconnect layer **400** will be described in the context of via interconnects.

Interconnect layer **400** can also include line interconnects, which are within the spirit and scope of this disclosure.

[0039] By way of example and not limitation, fabrication of interconnect layer **400** can be described as follows: etch stop layers **405** and **410** can be blanket deposited over interconnect layer **300**. Etch stop layer **405** can be SiCN or AlN with a thickness between about 10 Å and about 150 Å, and capping layer **410** can be aluminum oxide (AlO) with a thickness between about 10 Å and about 40 Å. ILD **415** can be subsequently formed over etch stop layer **410**. In some embodiments, ILD **415** can be a low-k dielectric material with a k-value below 3.9, silicon nitride, silicon oxide, silicon oxynitride, FSG, or UDOX. In some embodiments, ILD **415** can be a stack of dielectrics such as a low-k dielectric and another dielectric. The stack of dielectrics can include, for example, (i) a low-k dielectric (e.g., carbon doped silicon oxide) and a silicon carbide with nitrogen doping; (ii) a low-k dielectric and a silicon carbide with oxygen doping; (iii) a low-k dielectric with silicon nitride; or (iv) a low-k dielectric with silicon oxide. ILD **415** can be deposited by an APCVD process, a HDPCVD process, or a PECVD process. Subsequently, an antireflective coating (ARC) **420** can be formed over ILD **415**. Antireflective coating **420** can suppress ultra violet (UV) or extreme ultra violet (EUV) light reflections during a subsequent photolithography step and minimize undesirable generation of standing waves. Standing waves can increase the edge roughness of the resulting patterned structures. Antireflective coating **420** also forms a flat surface, on which a photoresist layer (not shown in FIG. 4) can be formed during a photolithography step, by operating as a “filler” that fills small imperfections on the top surface of ILD **415**. In some embodiments, antireflective coating **420** can be a nitrogen-free antireflective coating (NFARC).

[0040] One or more via openings **425** can be formed in ILD **415** by a photolithography process. The photolithography process can include, for example, the deposition and patterning of a photoresist layer (not shown in FIG. 4) over antireflective coating **420**, followed by an etch process that forms via openings **425** in ILD **415** in predetermined locations as shown in FIG. 4. In some embodiments, the formation of via openings **425** can be performed in two or more etch operations which can include different etch chemistries. The surfaces of via openings **425** can be subjected to a wet clean to remove byproducts of the etch process. In some embodiments, the aspect ratio (e.g., the ratio of height **425** to width **425_w**) for via opening **425** is between about 2:1 and about 6:1 (e.g., about 3:1). However, the aforementioned aspect ratio range is not limiting and more aggressive aspect ratios (e.g., about 7:1, about 8:1, about 10:1, etc.) can be used.

[0041] In FIG. 5, a barrier layer **500** can be blanket-deposited in via openings **425** to cover the exposed surfaces of via openings **425**. Further, barrier layer **500** can also cover the top surface of antireflective coating **420**. In some embodiments, barrier layer **500** can be a single layer or a stack of two or more layers. For example, barrier layer **500** can be a Co layer deposited by ALD or a TaN/Ta stack (e.g., TaN being the bottom layer and Ta being the top layer of the stack) deposited by PVD. Via openings **425** can be subsequently filled with metal fill **505**. Metal fill **505** is deposited over barrier layer **500** in via openings **425** and over antireflective coating **420**. In some embodiments, metal fill **505** can be Cu or a Cu alloy (e.g., CuMn). By way of example

and not limitation, metal fill **505** can be electroplated so that it fills via opening **425** without forming voids. A subsequent chemical mechanical planarization (CMP) process can be used to remove metal fill **505**, barrier layer **500**, and antireflective coating **420** over ILD **415** so that the top surfaces of ILD **415**, metal fill **505** and barrier layer **500** are substantially coplanar. The CMP process concludes the formation of vias **510**. Lines in interconnect layer **400** (not shown) can be formed concurrently with vias **510**. By way of example and not limitation, the size of vias **510** can be smaller, larger, or equal to the size of vias **305** in interconnect layer **300**.

[0042] Referring to FIG. 2, fabrication method **200** continues with operation **230** and the formation of a diffusion barrier layer on the one or more vias **510** of interconnect layer **400**. In some embodiments, the diffusion barrier of operation **230** is formed on a top surface of metal fill **505**. According to some embodiments, the formation of the diffusion barrier layer prevents the out-diffusion of Cu from metal fill **505**. For example, referring to FIG. 6, a diffusion barrier layer (or capping layer) **600** can be selectively formed on a top surface of metal fill **505** of vias **510**. In some embodiments, diffusion barrier layer or capping layer **600** can have a thickness that ranges from about 10 Å to about 100 Å. According to some embodiments, a thinner diffusion barrier layer **600** (e.g., with a thickness less than about 10 Å) may not prevent the out-diffusion of metal fill **505**, while a thicker diffusion barrier layer **600** (e.g., with a thickness greater than about 100 Å) may unnecessarily increase the deposition time of the diffusion barrier layer, which can in turn impact the deposition process throughput. Diffusion barrier layer **600** can include a metal, such as Co or Ru. In some embodiments, the diffusion barrier layer **600** is a polycrystalline material that does not chemically interact with metal fill **505** to form a compound. Additionally, diffusion barrier layer **600** can be deposited by a variety of deposition techniques including: physical vapor deposition (PVD), CVD, plasma-enhanced CVD (PECVD), atomic layer deposition (ALD), or plasma-enhanced ALD (PEALD). Diffusion barrier layer **600** can be deposited at a temperature between about 100° C. and about 500° C., according to some embodiments. The process pressure, during deposition, can range from about 0.1 Torr to about 100 Torr depending on the reactor geometry and the deposition technique.

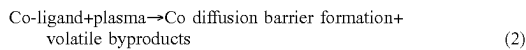
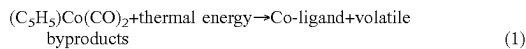
[0043] For example purposes, diffusion barrier layer or capping layer **600** will be described in the context of Co metal deposited with a PEALD process. Based on the disclosure herein, additional materials (e.g., Ru) and/or other deposition methods can be used. These additional materials and other deposition methods are within the spirit and scope of this disclosure.

[0044] In some embodiments, an organometallic precursor—for example, cyclopentadienylcobalt dicarbonyl ((C₅H₅)Co(CO)₂)—can be used to selectively deposit Co on metal fill **505**. Other Co organometallic precursors, such as cobalt (II) sulfate (CoSO₄), cobalt (II) nitrate (Co(NO₃)₂), or sodium cobaltinitrite can be used and are within the spirit and the scope of the present disclosure. The selective formation of Co diffusion barrier layer **600** is described below. The top surface of metal fill **505** is pre-treated with one or more gases that include, but are not limited to, argon (Ar), hydrogen (H₂), ammonia (NH₃) or any combination thereof. The aforementioned gas or gases can chemically

reduce (e.g., remove) a native oxide that is formed on the top surface of metal fill **505**. The thickness of the native oxide is less than about 100 Å. Oxidation of Cu (or CuMn) vias can occur, for example, during vacuum breaks between processing operations and can undesirably increase via/line resistance. The time period for the pre-treatment can be from about 10 s to about 30 s and can be performed at a temperature between about 100° C. and about 500° C., according to some embodiments.

[0045] Without a vacuum break, interconnect layer **400** is exposed to the Co precursor at a temperature between about 100° C. and about 500° C. In some embodiments, the formation of the diffusion barrier layer is a two-step process. During the first step, the Co precursor is partially thermally-decomposed over the exposed surfaces of interconnect layer **400**. In some embodiments, the partially decomposed Co precursor can be physisorbed (e.g., weakly bonded via electrostatic forces) on ILD **415** and chemisorbed (e.g., strongly bonded via chemical bonding) on the exposed surface of interconnect layer **400**. During a subsequent evacuation cycle, the physisorbed, and partially decomposed, Co precursor can be removed from ILD **415**. As a result, one or more monolayers of partially decomposed precursor can be favorably formed over exposed surfaces of metal fill **505**.

[0046] During the second step, a plasma can be used to fully decompose the one or more monolayers of the partially decomposed precursor to form Co diffusion barrier layer **600** on metal fill **505**. In some embodiments, volatile byproducts from the precursor's decomposition are concurrently removed from the reactor via evacuation (e.g., by pumping-down the reactor). By way of example and not limitation, in the case of a cyclopentadienyl cobalt dicarbonyl precursor, the chemical reactions can be described by the two following steps:



[0047] In some embodiments, the precursor exposure, evacuation cycles, and plasma exposures can be repeated until a desired thickness of Co is formed (e.g., between about 10 Å to about 100 Å). According to some embodiments, the plasma can be a mixture of one or more of the following gases: Ar, H₂, ozone (O₃), nitrogen (N₂), and/or NH₃. According to some embodiments, the plasma treatment can be performed at a temperature between about 100° C. and about 500° C.

[0048] In some embodiments, and as a result of the aforementioned deposition process, diffusion barrier layer **600** does not form over ILD **415**.

[0049] Referring to FIG. 2, fabrication method **200** continues with operation **240** and the formation of an MTJ structure over each of the Co diffusion barrier layers **600**. For example, as shown in FIG. 7, MTJ structures **700** can be formed in ILD **725** and over selected vias **510** of interconnect layer **400**. As discussed earlier, Co diffusion barrier layer **600** can prevent or reduce Cu out-diffusion from metal fill **505** to MTJ structures **700** by blocking mobile Cu atoms from reaching bottom electrode **710** and MTJ stack **715**. In some embodiments, interconnect layer **400** may have additional vias (e.g., like vias **510**) that may not be connected to

MTJ structures **700**. These vias, which are not connected to MTJ structures **700**, may not have a diffusion barrier layer **600** formed thereon.

[0050] In some embodiments, additional interconnect layers can be formed over top electrode **720** and ILD **725**, as discussed above in FIG. 1. For example, an interconnect layer, like interconnect layer **110** shown in FIG. 1, can be formed over MTJ structures **700** so that top electrodes **720** of MTJ structures **700** are in contact with respective vias of the interconnect layer.

[0051] FIG. 8 is a flowchart of an exemplary fabrication method **800** for forming a copper-free (Cu-free) interconnect layer on which MTJ structures can be formed. Since the interconnect layer is Cu-free (e.g., contains no Cu-based material), there are no Cu atoms available to diffuse to the MTJ structures. Fabrication method **800** may not be limited to the operations described below and other fabrication operations may be performed between the various operations of fabrication method **800** and are omitted merely for clarity.

[0052] In referring to FIG. 8, exemplary fabrication method **800** begins with operation **810** where one or more interconnect or contact layers are formed on a substrate. In some embodiments, the substrate with the one or more interconnect layers can be a partially fabricated wafer in BEOL, where the formed interconnect layers include—for example—Cu-based conductive structures (e.g., vias and lines). Additional layers can be formed between the substrate and the one or more interconnect layers, such as MOL and FEOL layers. An FEOL layer can include, for example, transistors, resistors, and capacitor structures, and an MOL layer can provide the electrical connections (e.g., contact structures) between the transistors in the FEOL layer and the one or more interconnect layers in the BEOL.

[0053] FIG. 3 is a cross-sectional view of an exemplary BEOL interconnect layer **300** which can serve as a starting point for fabrication method **800**, according to some embodiments. As discussed above, interconnect layer **300** can be, for example, a top interconnect layer from one or more BEOL interconnect layers over a substrate. In some embodiments, interconnect layer **300** is a Cu metallization layer. That is, metal fill **320** in vias **305** can include electroplated Cu or a Cu-alloy (e.g., CuMn). Interconnect layer **300** can also include lateral conductive structures (e.g., lines), which are not shown in FIG. 3 for simplicity. In the example of FIG. 3, the underlying BEOL interconnect layers, MOL layers, FEOL layers, and the substrate are not shown merely for clarity. In other words, BEOL interconnect layer **300** can be a top layer of a partially fabricated wafer, according to some embodiments.

[0054] In referring to FIG. 8, fabrication method **800** continues with operation **820** and the formation of a Cu-free interconnect layer over interconnect layer **300**. The fabrication process of the Cu-free interconnect can be described using FIG. 4. By way of example and not limitation, etch stop layers **405** and **410** can be blanket deposited over interconnect layer **300**. Etch stop layer **405** can be SiCN or AlN with a thickness between about 10 Å and about 150 Å, and capping layer **410** can be A **1203** with a thickness between about 10 Å and about 40 Å. ILD **415** can be subsequently formed over etch stop layer **410**. In some embodiments, ILD **415** can be a low-k dielectric material with a k-value below 3.9, silicon nitride, silicon oxide, silicon oxynitride, FSG, or UDOX. In some embodiments,

ILD 415 can be a stack of dielectrics such as a low-k dielectric and another dielectric. The stack of dielectrics can include, for example, (i) a low-k dielectric (e.g., carbon doped silicon oxide) and a silicon carbide with nitrogen doping; (ii) a low-k dielectric and a silicon carbide with oxygen doping; (iii) a low-k dielectric with silicon nitride; or (iv) a low-k dielectric with silicon oxide. ILD 415 can be deposited by an APCVD process, a HDPCVD process, or a PECVD process. Subsequently, an ARC 420 can be formed over ILD 415.

[0055] One or more via openings 425 can be formed in ILD 415 by a photolithography process. The photolithography process can include, for example, the deposition and patterning of a photoresist layer (not shown in FIG. 4) over antireflective coating 420, followed by an etch process that forms via openings 425 in ILD 415 in predetermined locations as shown in FIG. 4. In some embodiments, the formation of via openings 425 can be performed in two or more etch operations which can include different etch chemistries. The surfaces of via openings 425 can be subjected to a wet clean to remove byproducts of the etch process. As discussed above, the aspect ratio of via opening 425 is between about 2:1 and about 6:1 (e.g., about 3:1). However, this is not limiting and via openings with more aggressive aspect ratios (e.g., about 7:1, about 8:1, about 9:1, about 10:1, etc.) can be formed as long as the Cu-free process can fill the via openings without voids.

[0056] According to some embodiments, a pre-clean process can be used to remove the native CuO layer from vias 305 at the bottom each via openings 425. The pre-clean process can include one or more sequential operations. By way of example and not limitation, the pre-clean process can include a dry etch process with a hydrogen (H_2)/ammonia (NH_3)/nitrogen trifluoride (NF_3) plasma followed by a dry etch process with a nitrogen (N_2)/hydrogen (H_2) plasma. During the pre-clean process direct current (DC) and radio frequency (RF) power signals can be applied to the plasma. In some embodiments, the DC power can range from about 100 Watts to about 2000 Watts, and the RF power can range from about 50 Watts to about 500 Watts.

[0057] Subsequently, a Cu-free metallization process can be used to fill via openings 425. For example, in FIG. 9, a barrier layer 900 and Cu-free metal fill 910 can be used. In some embodiments, barrier layer 900 is a stack that includes a bottom layer of titanium (Ti) and a top layer of titanium nitride (TiN). The thickness of barrier layer 900 can range from about 10 Å to about 100 Å. In some embodiments, the TiN top layer of barrier layer 900 is doped with elements including fluorine (F), oxygen (O), nitrogen (N), chlorine (Cl), silicon (Si) carbon (C), arsenic (As), germanium (Ge), or cobalt (Co). In some embodiments, the TiN thickness can be adjusted so that the resistivity of the top TiN layer of barrier layer 900 is between about 10 $\mu\Omega$ -cm to about 200 $\mu\Omega$ -cm.

[0058] According to some embodiments, Ti can be blanket-deposited by PVD-based method (e.g., sputtering), and TiN can be deposited by an ALD or a PEALD process. In some embodiments, barrier layer 900 is deposited over antireflective coating 420 (shown in FIG. 4). By way of example and not limitation, Cu-free metal fill 910 can include W which may be blanket-deposited by CVD, ALD, or a combination thereof. For example, the W deposition can include an ALD-deposited W nucleation layer with a thickness between about 5 Å and about 100 Å, followed by a

CVD-deposited W fill with a thickness between about 200 nm to about 500 nm. By way of example and not limitation, the precursor for W can be organometallic (e.g., tungsten carbonyl) or halide-based (e.g., WF_6) with silane and/or hydrogen as the co-reactants. After the deposition of Cu-free metal fill 910, a CMP process can be used to remove metal fill 910, barrier layer 900 and antireflective coating 420 over ILD 415 so that the top surfaces of metal fill 910, barrier layer 900, and ILD 415 are coplanar, as shown in FIG. 9. After the CMP process, the formation of Cu-free vias 915 is complete.

[0059] In operation 830 of fabrication method 800 shown in FIG. 8, one or more MTJ structures 700 can be formed directly on Cu-free interconnect layer 400 of FIG. 9, as shown in FIG. 10. As discussed above, since interconnect layer 400 of FIG. 9 does not use Cu-based materials to form vias 915, there are no Cu atoms available to diffuse to MTJ structures 700 through bottom electrodes 710.

[0060] In some embodiments, MTJ structures 700 are limited to certain areas of the substrate, in which these certain areas do not include MTJ structures 700. For example, ILD 725 can include, adjacent to MTJ structures 700, additional conductive structures (e.g., vias and lines) where MTJ structures 700 are not present. These additional conductive structures are not shown in FIG. 7. In some embodiments, Cu-free vias 915 can be formed below MTJ structures 700 and Cu-based vias in interconnect layer 400 can be formed below other conductive structures of ILD 725. Therefore, interconnect layer 400 can also include Cu-based vias in these certain areas of the substrate, where MTJ structures 700 are not present.

[0061] In some embodiments, additional interconnect layers can be formed over top electrode 720 and ILD 725, as discussed above in FIG. 1. For example, an interconnect layer, like interconnect layer 110 shown in FIG. 1, can be formed over MTJ structures 700 so that top electrodes 720 of MTJ structures 700 are in contact with respective vias of the interconnect layer. In some embodiments, the conductive structures (e.g., lines and vias) of the interconnect layer formed over MTJ structures 700, can be filled with a Cu-based conductive material or a conductive material that is different from Cu-free metal fill 910.

[0062] The present disclosure is directed to exemplary interconnect fabrication methods that can prevent or reduce out-diffusion of Cu from interconnect layers to MTJ structures. According to some embodiments, a Co or Ru diffusion barrier layer can be formed between the Cu interconnects and the MTJ structures to prevent diffusion of Cu between the vias in the interconnect layer and the MTJ structure. The Co or Ru diffusion barrier layer can be selectively formed over the Cu interconnects. In another embodiment, a Cu-free interconnect layer can be formed using a W metallization scheme in place of Cu. W atoms, unlike Cu atoms, are not mobile and therefore do not out-diffuse to neighboring structures.

[0063] In some embodiments, a method includes forming an interconnect layer over a substrate, where forming the interconnect layer includes forming an interlayer dielectric stack with openings therein; disposing a metal in the openings to form corresponding conductive structures; and selectively depositing a diffusion barrier layer on the metal. In the method, selectively depositing the diffusion barrier layer includes pre-treating the surface of the metal; disposing a precursor to selectively form a partially-decomposed pre-

cursor layer on the metal, and exposing the partially-decomposed precursor layer to a plasma to form the diffusion barrier layer. The method further includes forming an MTJ structure on the interconnect layer over the diffusion barrier layer, where the bottom electrode of the MTJ structure is aligned to the diffusion barrier layer.

[0064] In some embodiments, a structure includes an interconnect layer disposed over a substrate, where the interconnect layer comprises a conductive structure filled with a conductive material; a diffusion barrier layer that prevents out-diffusion of the conductive material and is disposed on the conductive structure; and a MTJ structure disposed on the interconnect layer, where the diffusion barrier layer is interposed between the bottom electrode of the MTJ structure and the conductive structure

[0065] In some embodiments, a structure includes a first interconnect layer disposed over a substrate, where the first interconnect layer includes one or more first conductive structures with a first conductive material; and a second interconnect layer that is disposed over the first interconnect layer, where the second interconnect layer includes one or more second conductive structures, in contact with the one or more first conductive structures, having a second conductive material. Further, the second conductive material is different from the first conductive material. The structure also includes one or more MTJ structures disposed on the second interconnect layer and in contact with the one or more second conductive structures, respectively, where the bottom electrodes of the one or more MTJ structures are in contact with the second conductive material of the one or more second conductive structures.

[0066] The foregoing outlines features of embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A structure, comprising:
a substrate;
an interconnect layer, disposed on the substrate, comprising:
a first conductive layer disposed on the substrate,
a doped nitride layer disposed on the first conductive layer, and
a second conductive layer disposed on the doped nitride layer; and
a magnetic tunnel junction (MTJ) structure disposed on the second conductive layer.
2. The structure of claim 1, wherein the first conductive layer comprises a titanium layer.
3. The structure of claim 1, wherein the second conductive layer comprises a copper-free metal layer.

4. The structure of claim 1, wherein the first and second conductive layers comprise metals different from each other.

5. The structure of claim 1, wherein the second conductive layer comprises a tungsten layer.

6. The structure of claim 1, wherein the doped nitride layer comprises a doped titanium nitride layer.

7. The structure of claim 1, wherein the doped nitride layer comprises fluorine, oxygen, nitrogen, chlorine, silicon, carbon, arsenic, germanium, or cobalt dopants.

8. The structure of claim 1, wherein the doped nitride layer is in contact with a bottom surface and sidewalls of the second conductive layer.

9. The structure of claim 1, wherein a bottom surface of the MTJ structure is in contact with top surfaces of the first conductive layer and the doped nitride layer.

10. The structure of claim 1, wherein the interconnect layer further comprises an etch stop layer in contact with sidewalls of the first conductive layer.

11. A structure, comprising:

a substrate;

a first via structure, disposed on the substrate, comprising a first metal layer;

a second via structure, comprising:

a second metal layer disposed on the substrate,

a metal nitride layer disposed on the second metal layer, and

a third metal layer disposed on the metal nitride layer, wherein the first, second, and third metal layers are different from each other; and

a magnetic tunnel junction (MTJ) structure disposed on the third metal layer.

12. The structure of claim 11, wherein the first metal layer comprises a copper-based layer.

13. The structure of claim 11, wherein the second metal layer comprises a titanium layer.

14. The structure of claim 11, wherein the third metal layer comprises a copper-free metal layer.

15. The structure of claim 11, wherein the metal nitride layer comprises dopants.

16. The structure of claim 11, further comprising an etch stop layer disposed on a top surface of the first via structure and in contact with sidewalls of the second via structure.

17. A method, comprising:

forming an interconnect structure on a substrate;

depositing a first metal layer on the interconnect structure;

depositing a metal nitride layer on the first metal layer;

depositing a second metal layer on the metal nitride layer, wherein the first and second metal layers comprise metals different from each other; and

forming a magnetic tunnel junction (MTJ) structure on the second metal layer and the metal nitride layer.

18. The method of claim 17, wherein depositing the second metal layer comprises depositing a copper-free metal.

19. The method of claim 17, wherein depositing the second metal layer comprises depositing a tungsten layer.

20. The method of claim 17, further comprising doping the metal nitride layer with metal dopants.

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